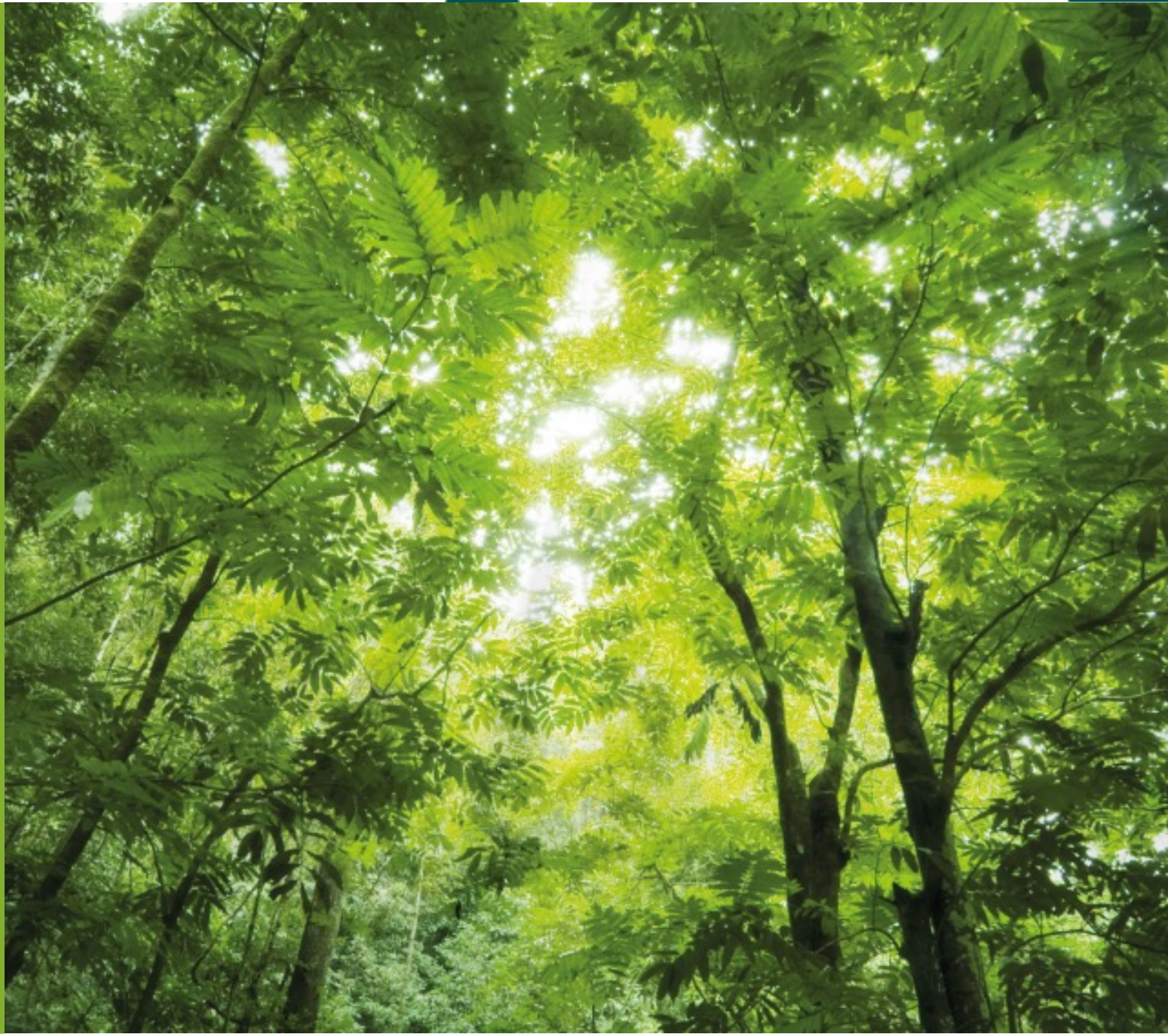
CHAPTER 6

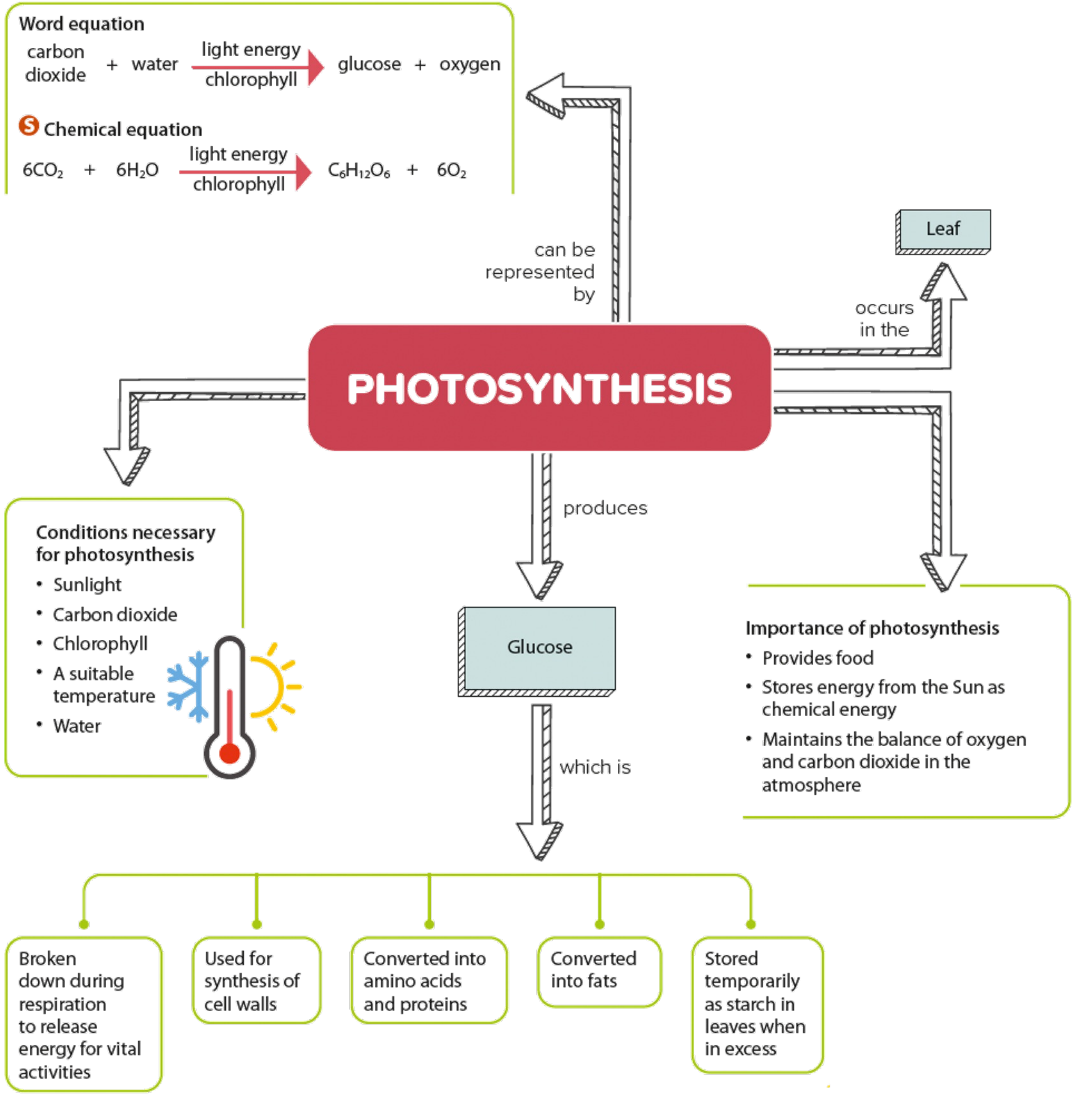
PLANT NUTRITION



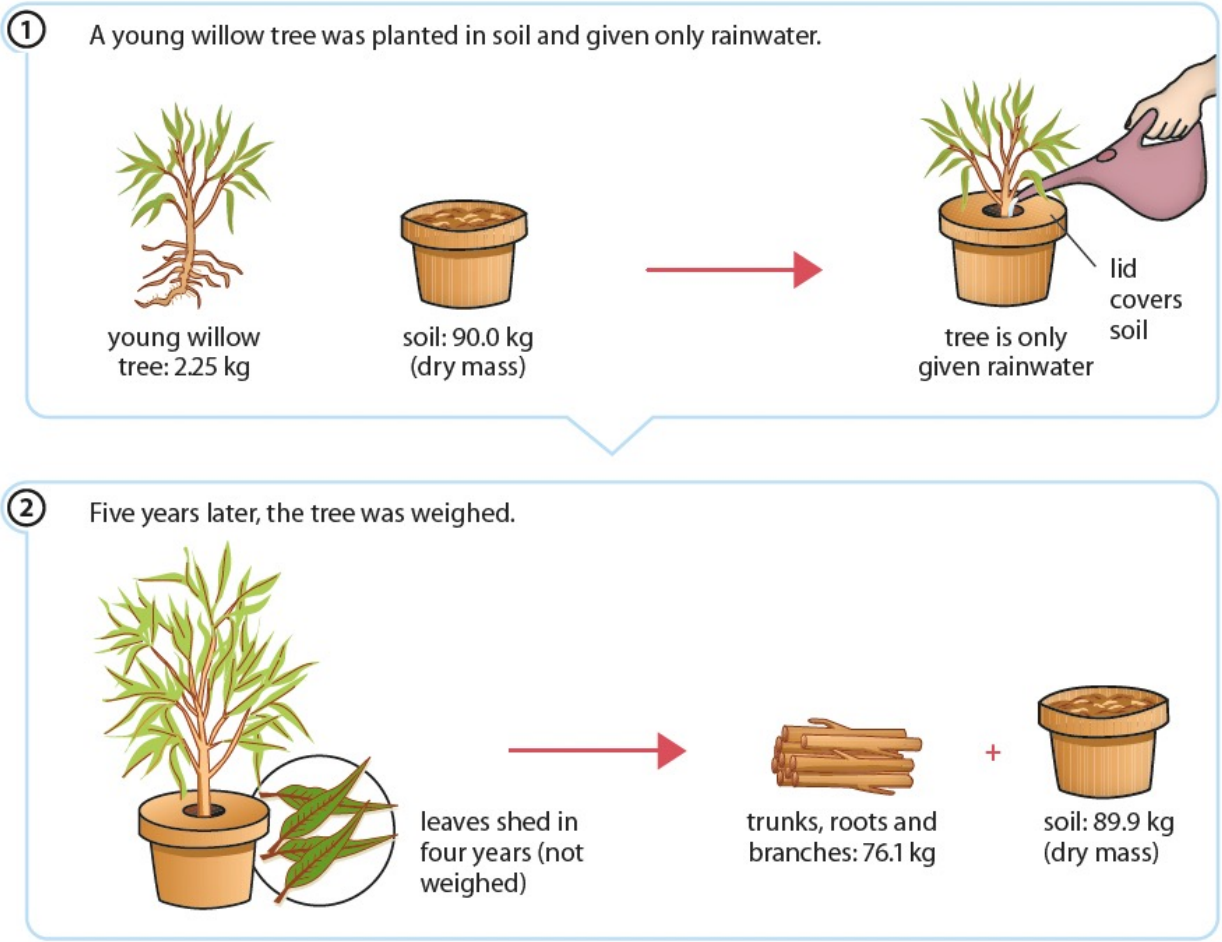
6.1 Photosynthesis

In this section, you will learn the following:

- Investigate the need for chlorophyll, light and carbon dioxide for photosynthesis, using appropriate controls.
- State that chlorophyll is a green pigment that is found in chloroplasts.
- Describe photosynthesis as the process by which plants synthesise carbohydrates from raw materials using energy from light.
- State that chlorophyll transforms light energy into chemical energy for the synthesis of carbohydrates.
- State the word equation for photosynthesis.
- S State the balanced chemical equation for photosynthesis.
- S Identify and explain the limiting factors of photosynthesis in different environmental conditions.
- Investigate and describe the effects of varying light intensity, carbon dioxide concentration and temperature on the rate of photosynthesis.
- Investigate the effect of light and dark conditions on gas exchange of an aquatic plant using hydrogencarbonate indicator solution.
- Outline the subsequent use and storage of the carbohydrates made in photosynthesis.
- Explain the importance of nitrate ions for making amino acids and magnesium ions for making chlorophyll.



Where do plants get their food?



Jan-Baptista van Helmont's willow experiment (Figure 6.2 of Student's Book)

- The experiment proved that plants do not get their food from the soil.
- Water and mineral ions do not provide carbon. Yet, carbon-rich organic molecules such as carbohydrates are found in plants.

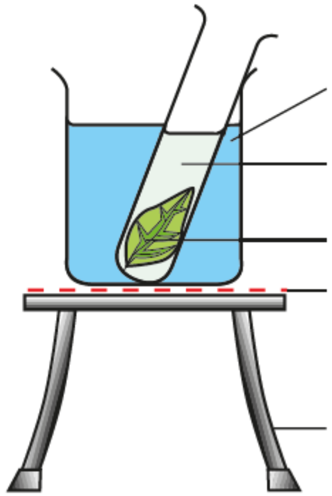
Where do the carbon and energy in plants come from?

How can we study photosynthesis?

- Green plants get their carbon and energy through a process called photosynthesis.
- You can design experiments to find out what happens in photosynthesis. First, you need to have the basic knowledge:

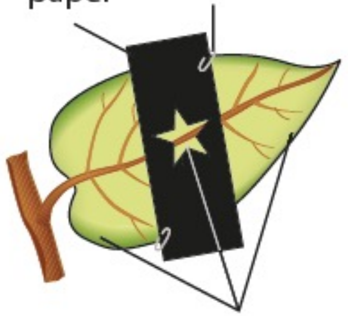
Basic knowledge
Glucose is first formed from carbon dioxide during photosynthesis.
Presence of starch in the leaves suggests that photosynthesis has taken place.
Destarching (removal of starch) must be carried out on the plants before the experiments.

Let's Investigate 6A



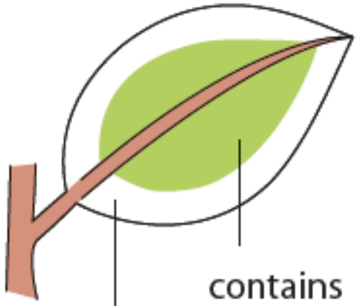
- The leaf becomes colourless after ten minutes.
- Iodine solution turns blue-black when added to the leaf, indicating that starch is present. This suggests that photosynthesis has taken place.

Let's Investigate 6B



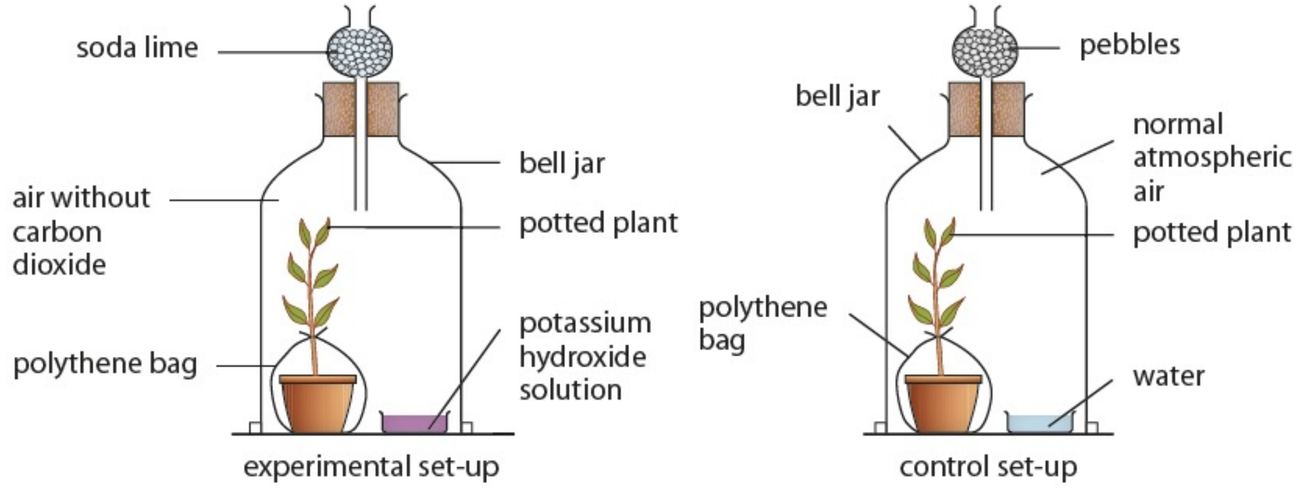
- The leaf is tested for starch after a few hours.
- Only the parts exposed to sunlight are stained blue-black, indicating that starch is present. This suggests that sunlight is necessary for photosynthesis.

Let's Investigate 6C



- The green parts of the leaf contain chlorophyll.
- When the leaf is tested for starch, only those parts containing chlorophyll are stained blue-black, indicating that starch is present. This shows that chlorophyll is necessary for photosynthesis.

Let's Investigate 6D



- The leaves from both plants are tested for starch after a few hours.
- The iodine remains brown for the leaf exposed to air without carbon dioxide. The iodine turns blue-black for the leaf in the control set-up. This shows that carbon dioxide is necessary for photosynthesis.

What conditions are essential for photosynthesis?

1. Sunlight

2. Carbon dioxide

3. Chlorophyll

4. A suitable temperature

5. Water

What happens during photosynthesis?

- **Photosynthesis** is the process where plants synthesise carbohydrates from water and carbon dioxide.
- Water and carbon dioxide are raw materials for photosynthesis.
- Oxygen is released during the process.
- The overall process can be summarised using this word equation:

$$\text{carbon dioxide} + \text{water} \xrightarrow[\text{chlorophyll}]{\text{light energy}} \text{glucose} + \text{oxygen}$$

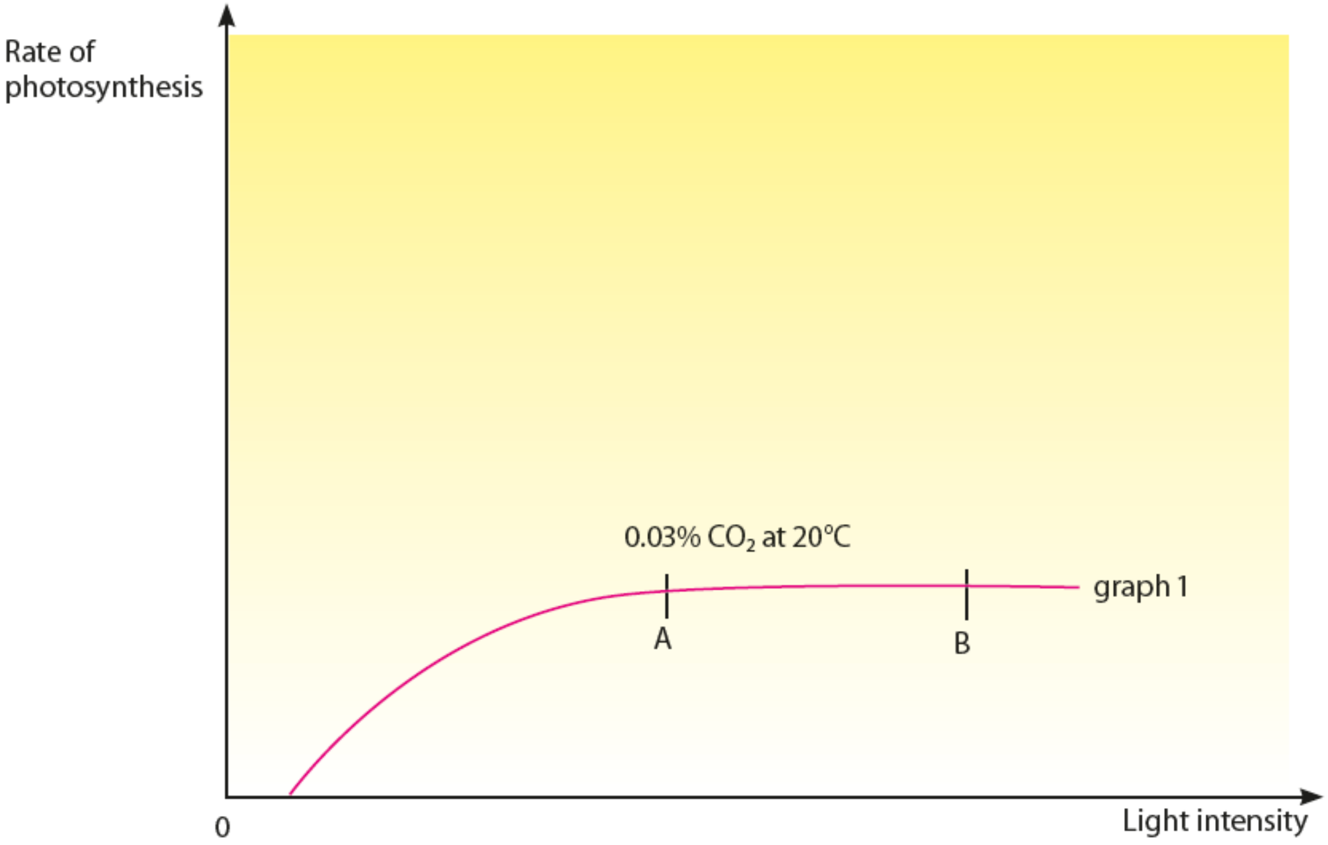
S

- The balanced chemical equation can be written as follows:

$$6CO_2 + 6H_2O \xrightarrow[\text{chlorophyll}]{\text{light energy}} C_6H_{12}O_6 + 6O_2$$

What are the limiting factors in photosynthesis?

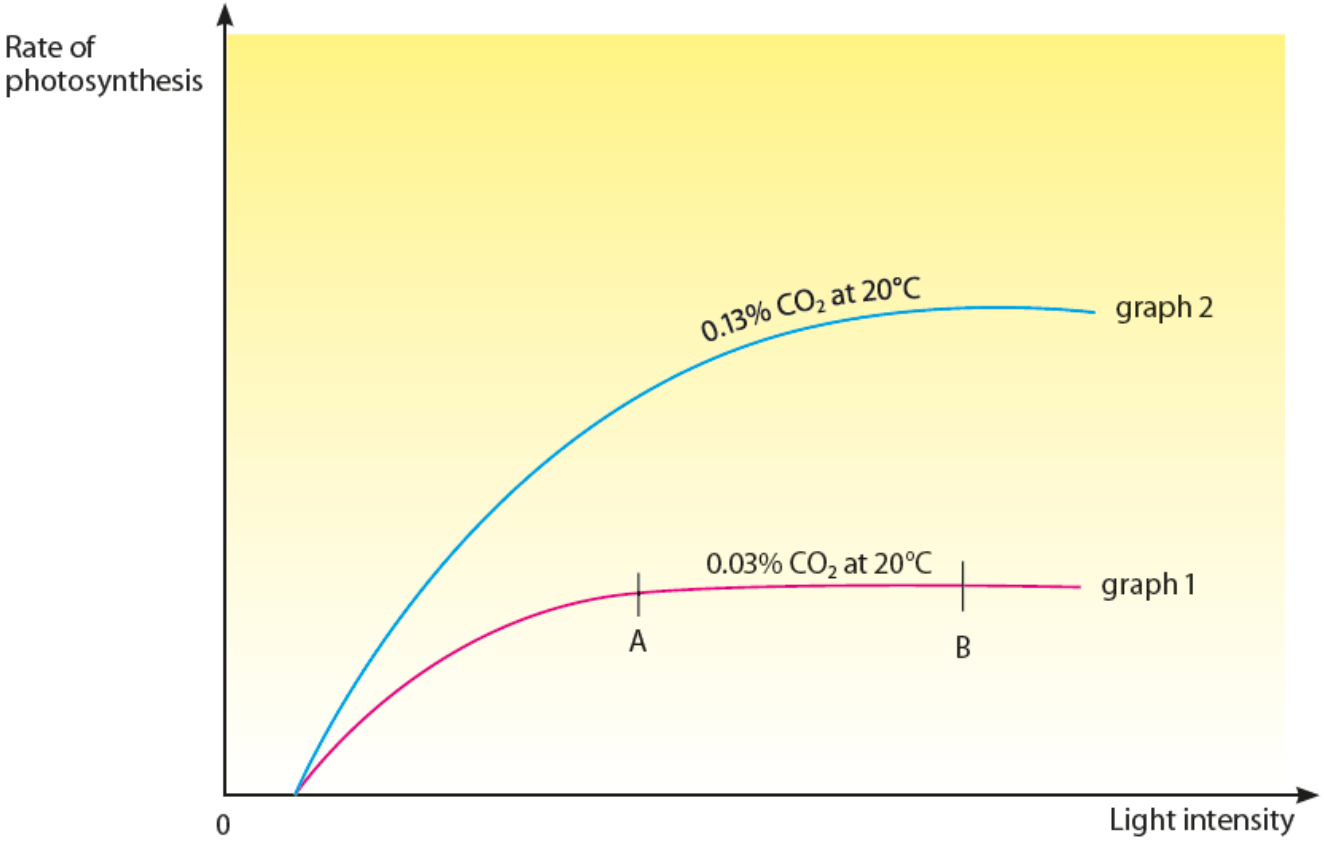
1. Light intensity



Effect of increasing light intensity on the rate of photosynthesis at 0.03% CO_2 at 20°C

What are the limiting factors in photosynthesis?

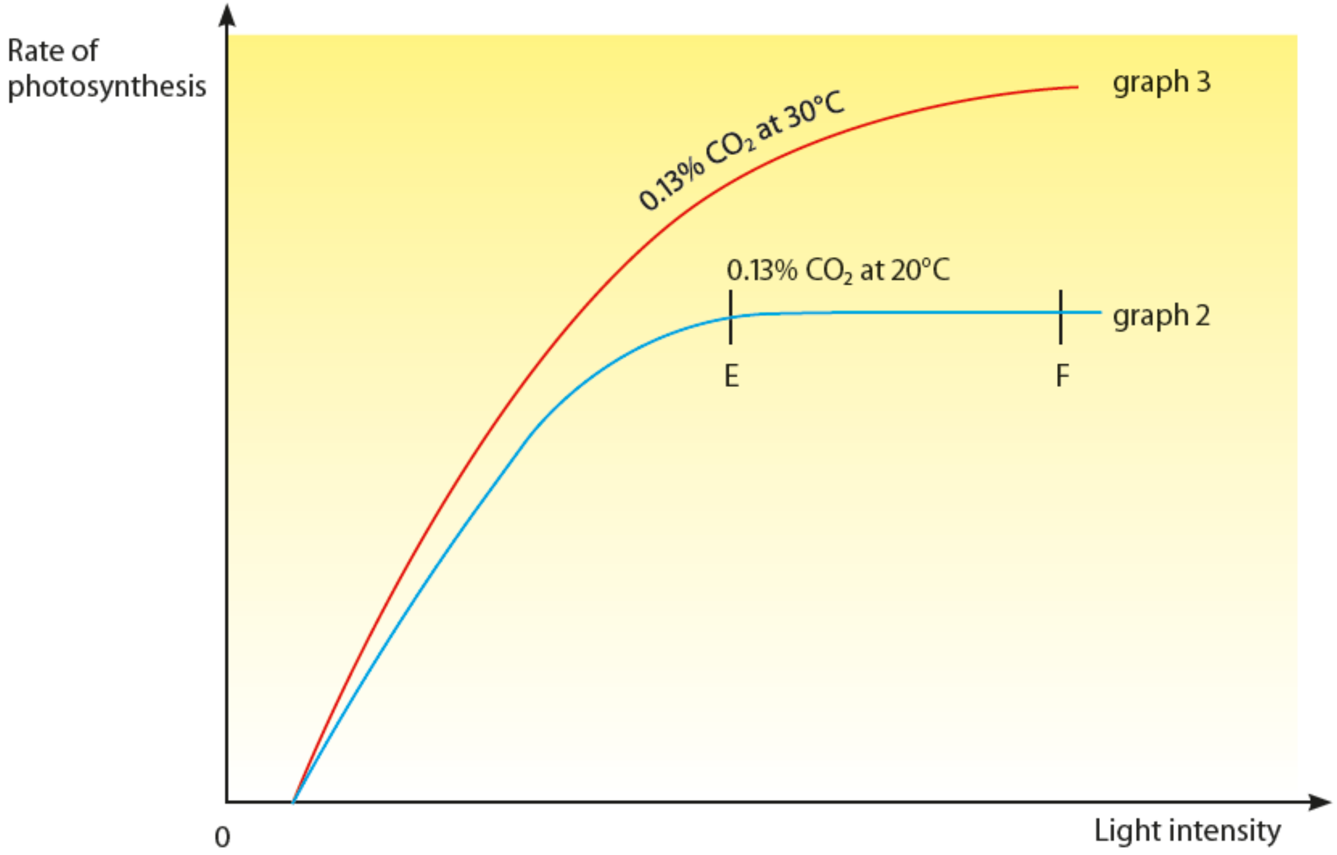
2. Carbon dioxide (CO_2) concentration



Effect of increasing carbon dioxide (CO_2) concentration on the rate of photosynthesis at 20°C

What are the limiting factors in photosynthesis?

3. Temperature

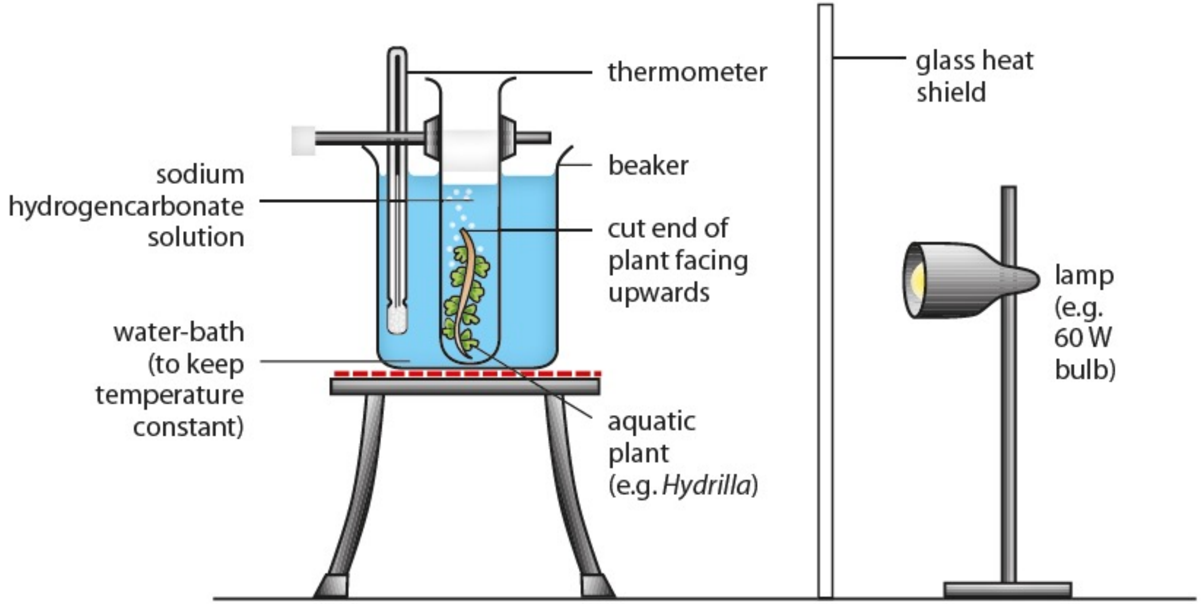


Effect of increasing temperature on the rate of photosynthesis at 0.13% CO_2

What is the effect of varying light intensity on the rate of photosynthesis?

Let's Investigate 6F

- The oxygen produced is measured as bubbles produced by the plant.
- Fewer bubbles are produced as the distance between the lamp and the plant increases.
- The higher the rate of bubbles given off, the higher the rate of photosynthesis.
- Thus, increasing the light intensity increases the rate of bubbles given off and therefore the rate of photosynthesis.



What is the effect of varying temperature on the rate of photosynthesis?

Let's Investigate 6G

- The oxygen produced is measured as bubbles produced by the plant.
- More bubbles are produced as the temperature increases from 5ºC to 45ºC.
- The higher the rate of bubbles given off, the higher the rate of photosynthesis.
- Thus, increasing the temperature increases the rate of bubbles given off and therefore the rate of photosynthesis.

What is the effect of varying carbon dioxide concentration on the rate of photosynthesis?

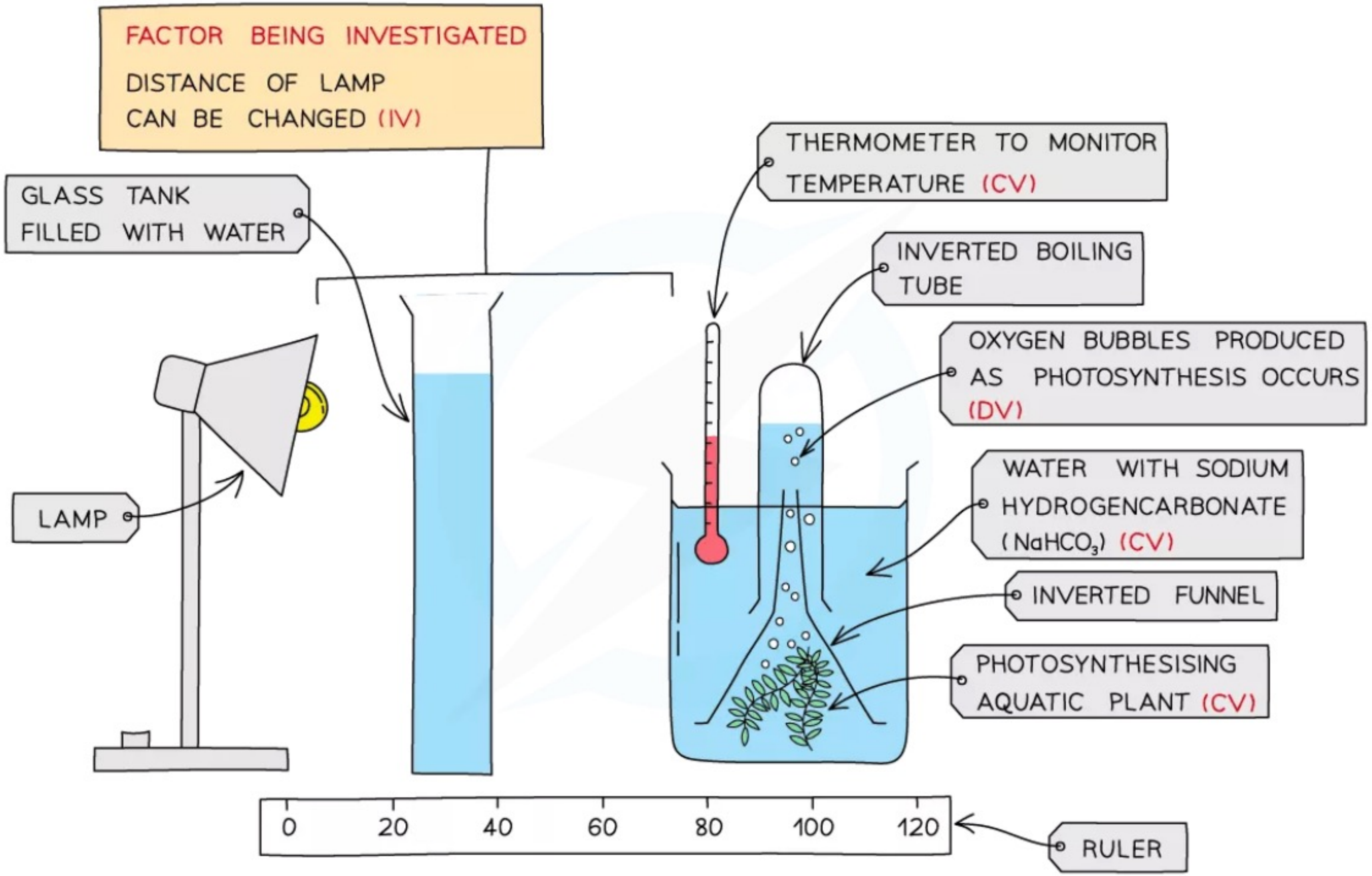
Let's Investigate 6H

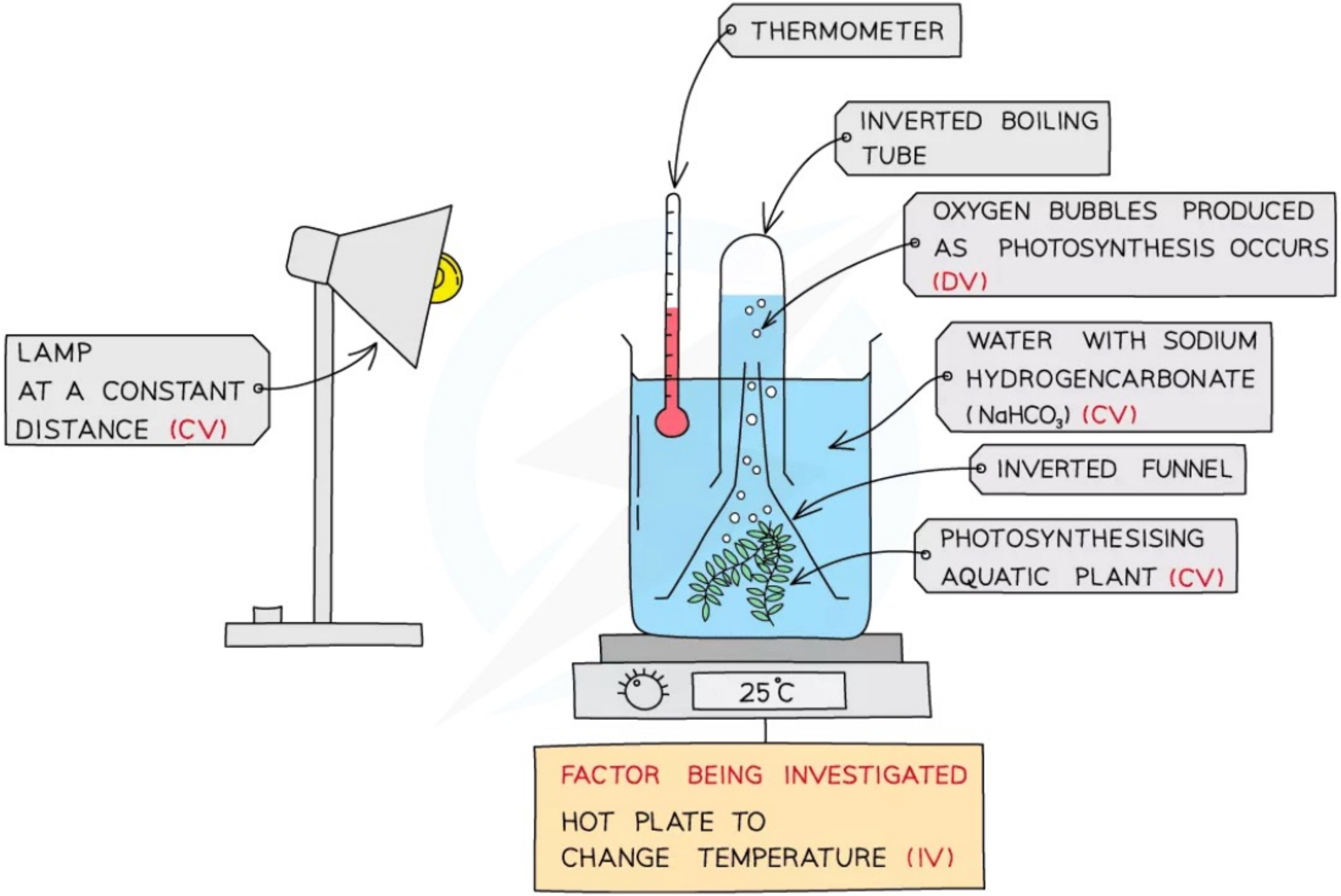
- The concentrations of sodium hydrogencarbonate solutions are proportional to the carbon dioxide concentration in the solution.
- More bubbles are produced as the concentration in sodium hydrogencarbonate solution increases.
- The higher the rate of bubbles given off, the higher the rate of photosynthesis.
- Thus, increasing the concentration of sodium hydrogencarbonate solution increases the rate of bubbles given off and therefore the rate of photosynthesis.

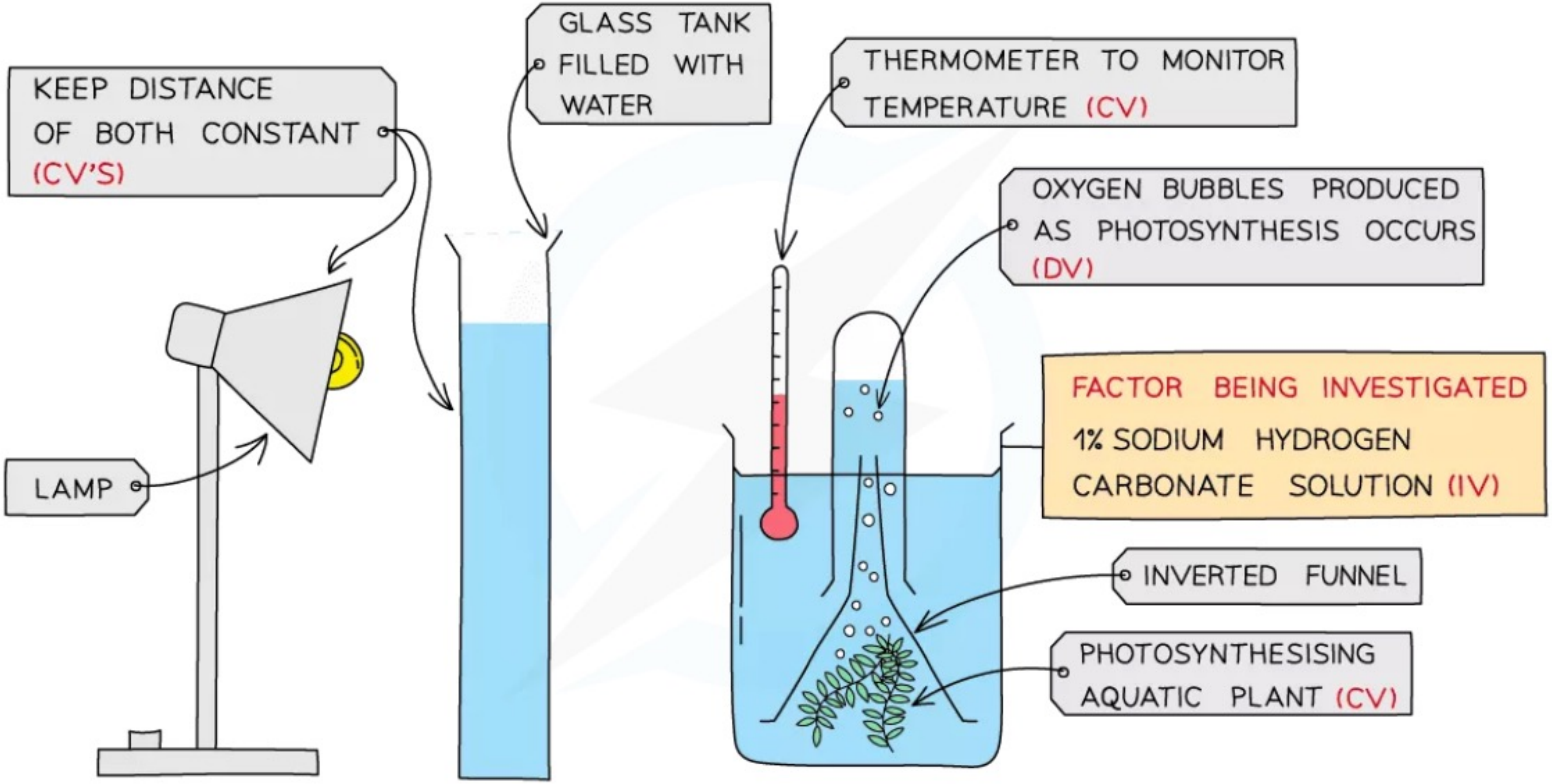
Investigating the Rate of Photosynthesis

- The plants usually used are ***Elodea*** or ***Camboba*** – types of pondweed
- As photosynthesis occurs, oxygen gas produced is released
- As the plant is in water, the oxygen released can be seen as **bubbles** leaving the cut end of the pondweed
- The number of **bubbles produced over a minute** can be counted to record the rate
- The more bubbles produced per minute, the faster the rate of photosynthesis
- A more accurate version of this experiment is to collect the oxygen released in a test tube inverted over the top of the pondweed over a longer period of time and then measure the **volume of oxygen** collected
- This practical can be used in the following ways:

Investigating the effect of **changing light intensity,** by moving a lamp different distances away from the beaker containing the pondweed:


FACTOR BEING INVESTIGATED
DISTANCE OF LAMP CAN BE CHANGED (IV)
GLASS TANK FILLED WITH WATER
LAMP
THERMOMETER TO MONITOR TEMPERATURE (CV)
INVERTED BOILING TUBE
OXYGEN BUBBLES PRODUCED AS PHOTOSYNTHESIS OCCURS (DV)
WATER WITH SODIUM HYDROGENCARBONATE ($NaHCO_3$) (CV)
INVERTED FUNNEL
PHOTOSYNTHESISING AQUATIC PLANT (CV)
0 20 40 60 80 100 120
RULER


THERMOMETER
INVERTED BOILING TUBE
OXYGEN BUBBLES PRODUCED AS PHOTOSYNTHESIS OCCURS (DV)
LAMP AT A CONSTANT DISTANCE (CV)
WATER WITH SODIUM HYDROGENCARBONATE ($NaHCO_3$) (CV)
INVERTED FUNNEL
PHOTOSYNTHESISING AQUATIC PLANT (CV)
25°C
FACTOR BEING INVESTIGATED
HOT PLATE TO CHANGE TEMPERATURE (IV)

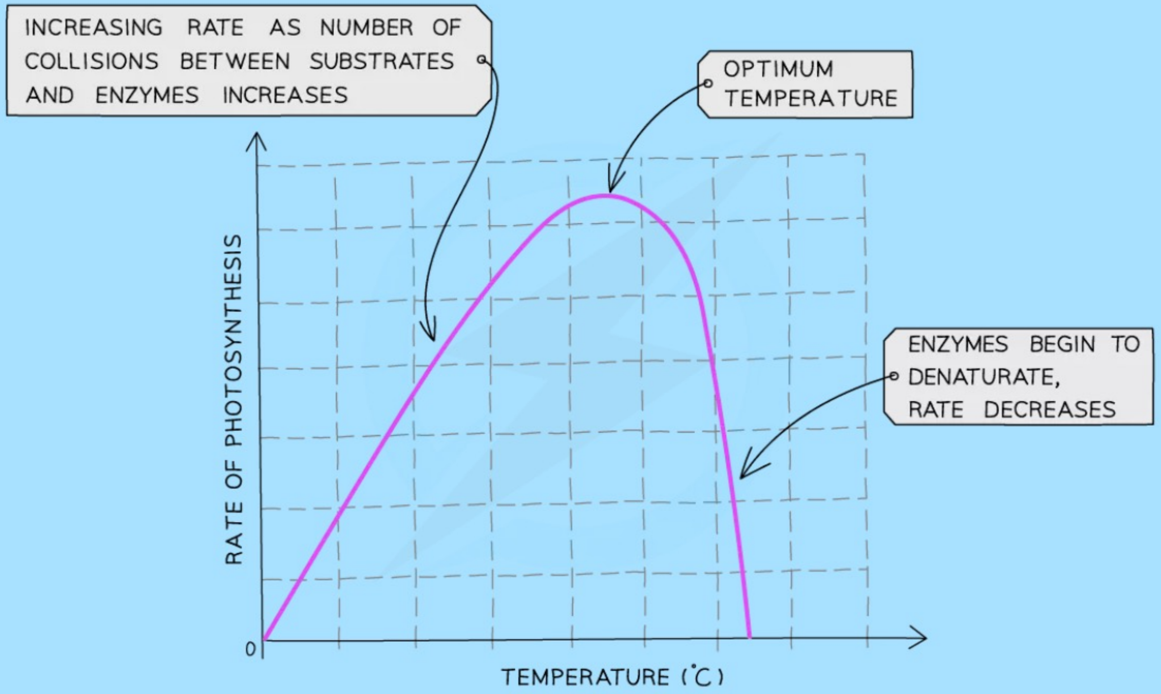

KEEP DISTANCE OF BOTH CONSTANT (CV'S)
GLASS TANK FILLED WITH WATER
THERMOMETER TO MONITOR TEMPERATURE (CV)
OXYGEN BUBBLES PRODUCED AS PHOTOSYNTHESIS OCCURS (DV)
FACTOR BEING INVESTIGATED
1% SODIUM HYDROGEN CARBONATE SOLUTION (IV)
LAMP
INVERTED FUNNEL
PHOTOSYNTHESISING AQUATIC PLANT (CV)

What is a Limiting Factor?

- If a plant is given unlimited sunlight, carbon dioxide and water and is at a warm temperature, the limit on the rate (speed) at which it can photosynthesise is its own ability to absorb these materials and make them react
- However, most often plants do not have unlimited supplies of their raw materials so their rate of photosynthesis is **limited** by **whatever factor is the lowest at that time**
- So a **limiting factor** can be defined as **something present in the environment in such short supply that it restricts life processes**
- There are **three** main factors which limit the rate of photosynthesis:
  - **Temperature**
  - **Light intensity**
  - **Carbon dioxide concentration**
- Although water is necessary for photosynthesis, it is **not considered a limiting factor** as the amount needed is relatively small compared to the amount of water transpired from a plant so there is hardly ever a situation where there is not enough water for photosynthesis

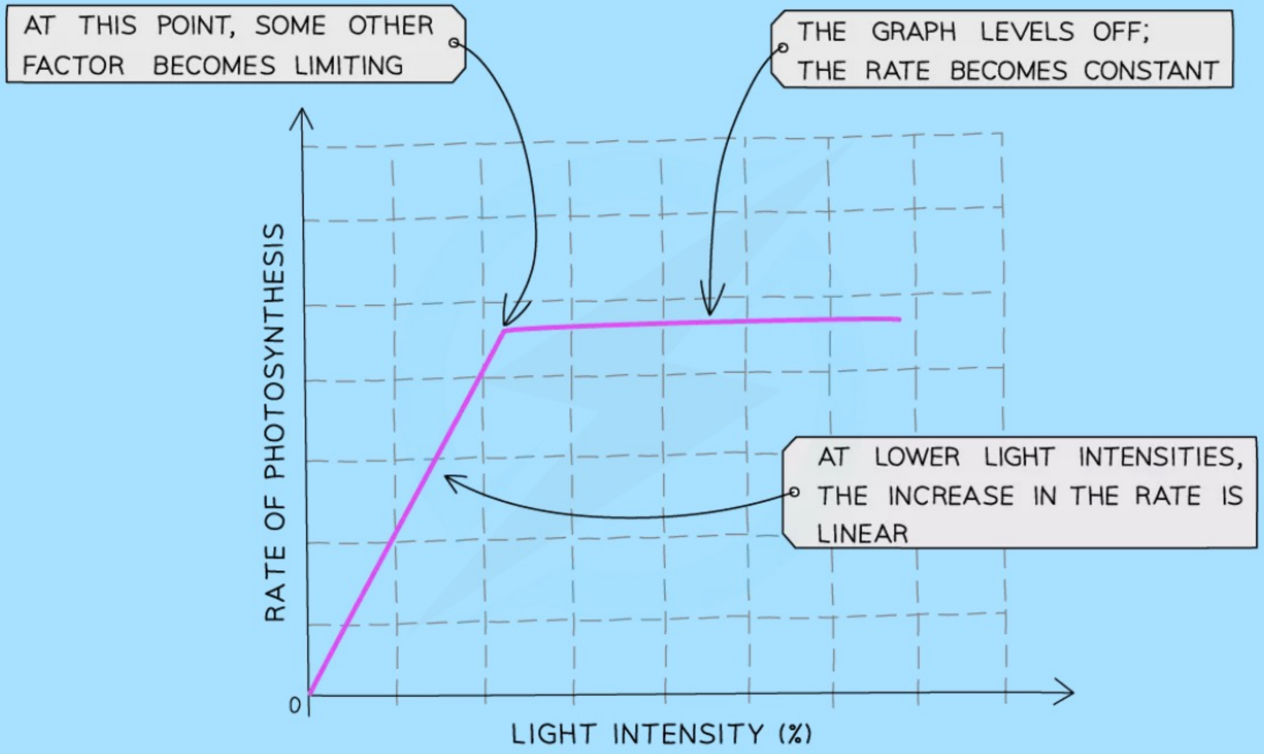
Temperature

- As temperature increases the rate of photosynthesis increases as the reaction is **controlled by enzymes**
- However, as the reaction is controlled by enzymes, this trend only continues up to a certain temperature beyond which the enzymes begin to **denature** and the rate of reaction **decreases**



Light intensity

- The **more light** a plant receives, the **faster the rate** of photosynthesis
- This trend will continue until some other factor required for photosynthesis prevents the rate from increasing further because it is now in short supply

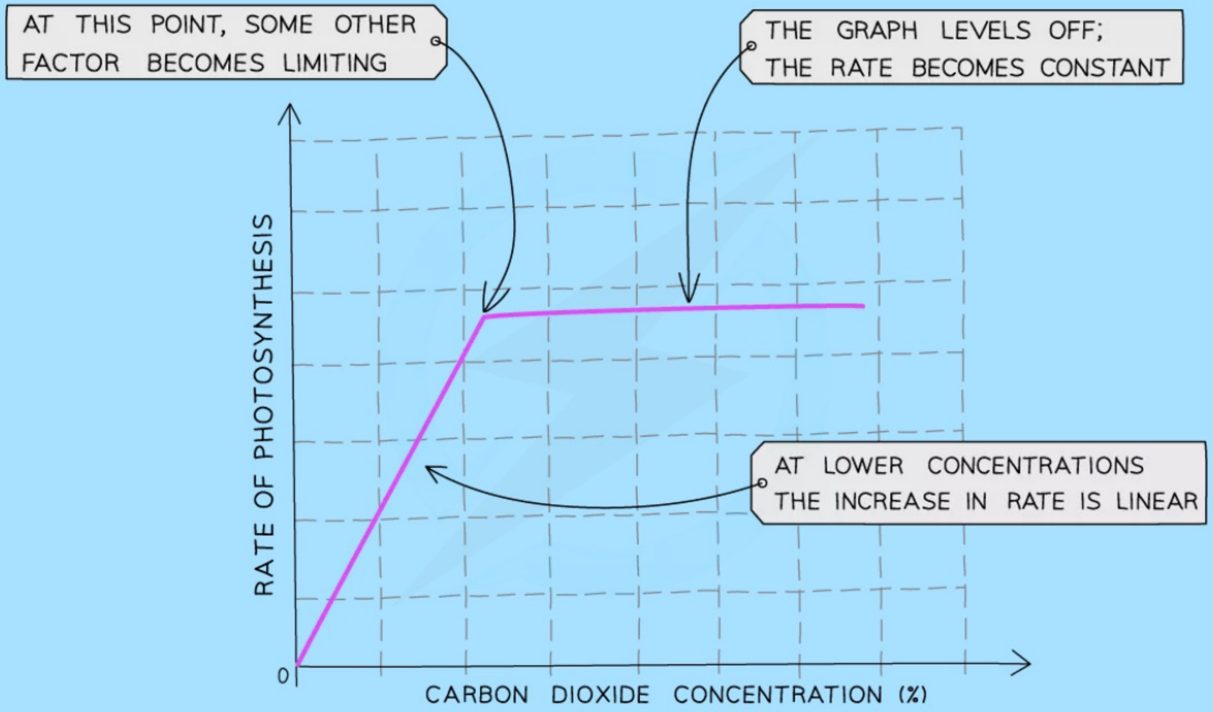


At low light intensities, increasing the intensity will initially increase the rate of photosynthesis. At a certain point, increasing the light intensity stops increasing the rate. The rate becomes constant regardless of how much light intensity increases as something else is limiting the rate.

- The factors which could be limiting the rate when the line on the graph is horizontal include **temperature not being high enough** or **not enough carbon dioxide.**

Carbon Dioxide Concentration

- Carbon dioxide is one of the raw materials required for photosynthesis
- This means the **more carbon dioxide** that is present, the **faster the reaction** can occur
- This trend will continue until some other factor required for photosynthesis prevents the rate from increasing further because it is now in short supply

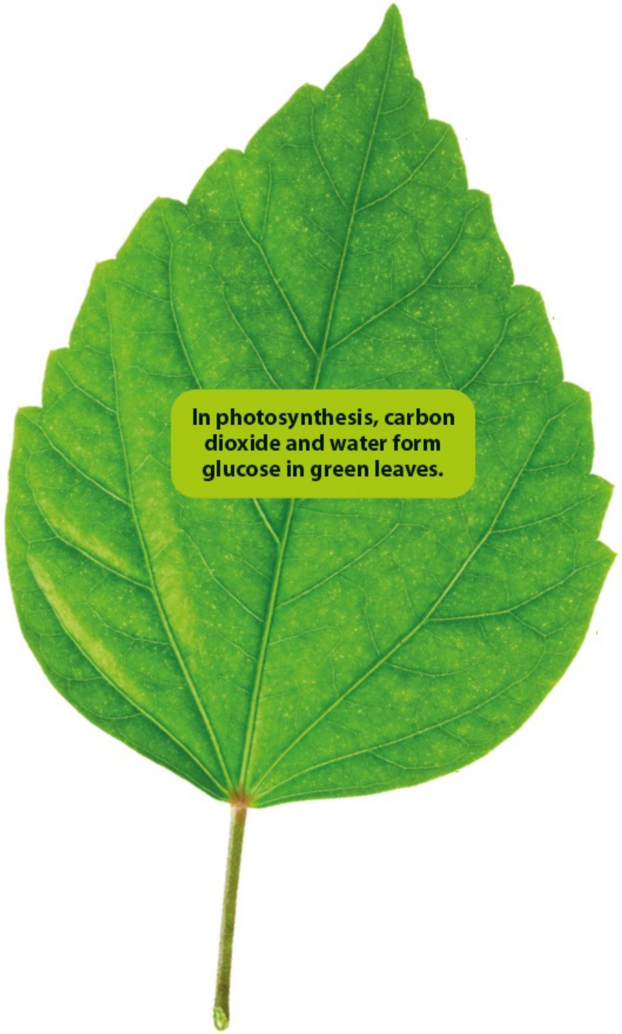


Changing Glasshouse Condition

- The knowledge about limiting factors and how they affect the rate of photosynthesis can be used to help **control factors in glass houses** to ensure **maximum crop yields** for farmers
- Growing crops outside does not allow farmers to control any of these factors to increase growth of plants
- In a glass house, several conditions can be manipulated to increase the rate of photosynthesis, including:
    - **artificial heating** (enzymes controlling photosynthesis can work faster at slightly higher temperatures – only used in temperature countries such as the UK)
    - **artificial lighting** (plants can photosynthesise for longer)
    - **increasing carbon dioxide content** of the air inside (plants can photosynthesise quicker)
    - **regular watering**

- When considering the use of glasshouses and manipulating conditions like this, farmers need to balance the **extra cost** of providing heating, lighting and carbon dioxide against the **increased income**
- In **tropical countries** where temperatures are much hotter, glasshouses may still be used to control other conditions however they may need to be **ventilated** to release hot air and **avoid temperatures rising too high**, which could cause the denaturation of the enzymes controlling the photosynthesis reaction

What happens to the glucose that is formed during photosynthesis?

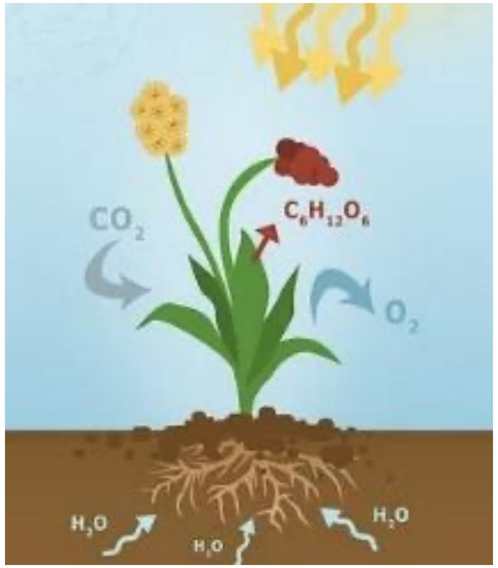


1. Glucose is used immediately.
2. In daylight, glucose forms faster than it can be removed. The excess glucose is converted to starch in leaves. In darkness, starch is converted back to glucose.
3. Glucose is converted to sucrose which
    - is transported for storage;
    - Is converted to starch;
    - may be converted back to glucose;
    - becomes a component of nectar in flowers.
4. Glucose reacts with nitrates and magnesium ions to form chlorophyll.
5. Glucose reacts with nitrates and other mineral ions to form amino acids.
6. Glucose forms fats for storage, use in cellular respiration and synthesis of new protoplasm.

Why is photosynthesis important?

- Photosynthesis makes chemical energy available to animals and other organisms.

- Photosynthesis removes carbon dioxide and provides oxygen.



- Energy is stored in fossil fuels through photosynthesis.

LINK

Carbon dioxide is constantly removed from and released into the environment in the carbon cycle. Find out more in Chapter 20.

Let's Practise 6.1

A few drops of iodine solution were added to two decolourised leaves, A and B. The results of each leaf are shown in Figure 6.16.

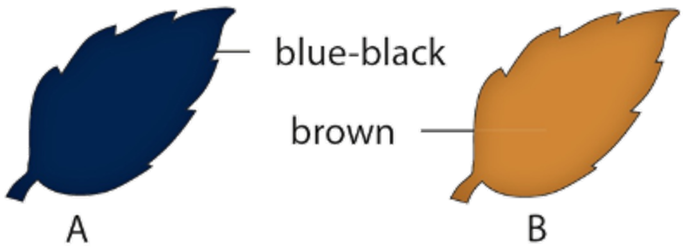


Figure 6.26 of Student's Book

a) What treatment do you think the leaves were given before they were plucked from the plants?

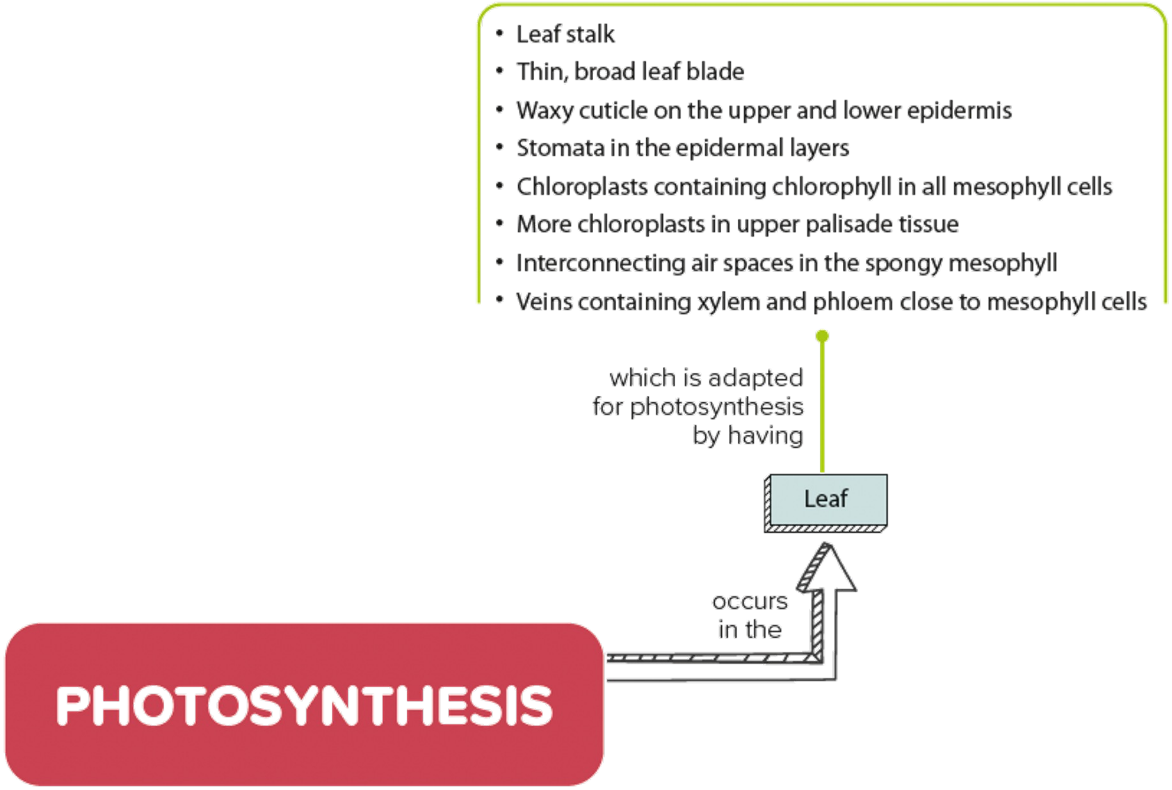
b) The experiment could have been used to show the need for a factor essential for photosynthesis. Name **one** such possible factor.

c) Which leaf, A or B, could serve as a control for the experiment in (b)? Give your reason.

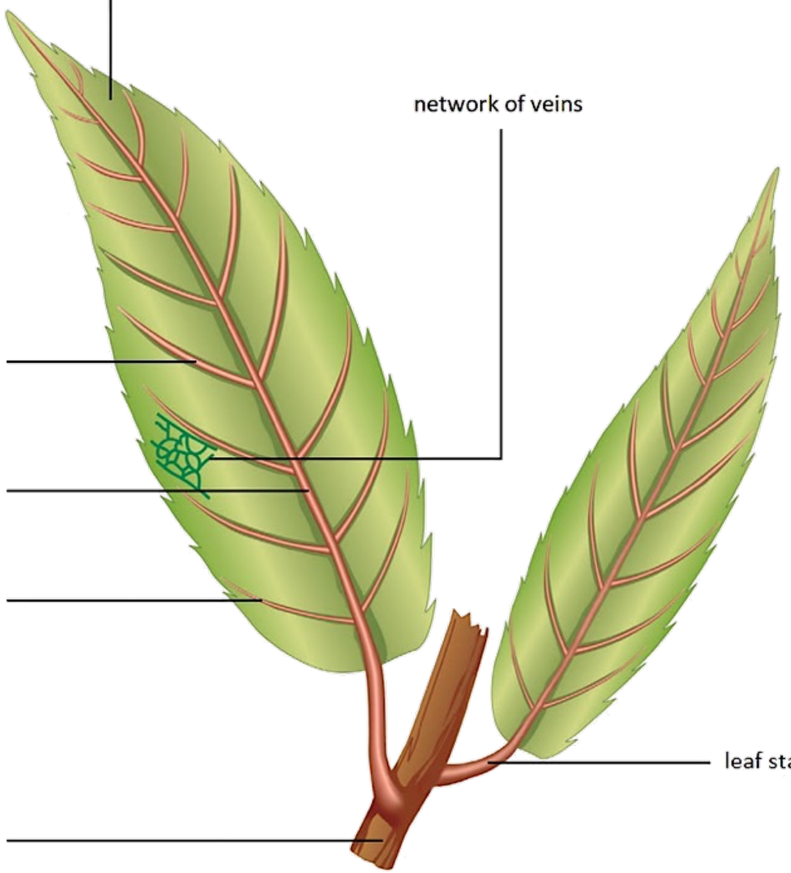
6.2 Leaf Structure and Function

In this section, you will learn the following:

- State that most leaves have a large surface area and are thin, and explain how these features are adaptations for photosynthesis.
- Identify in diagrams and images the various structures in leaves of a dicotyledonous plant.
- Explain how the structures make leaves adapted for photosynthesis.



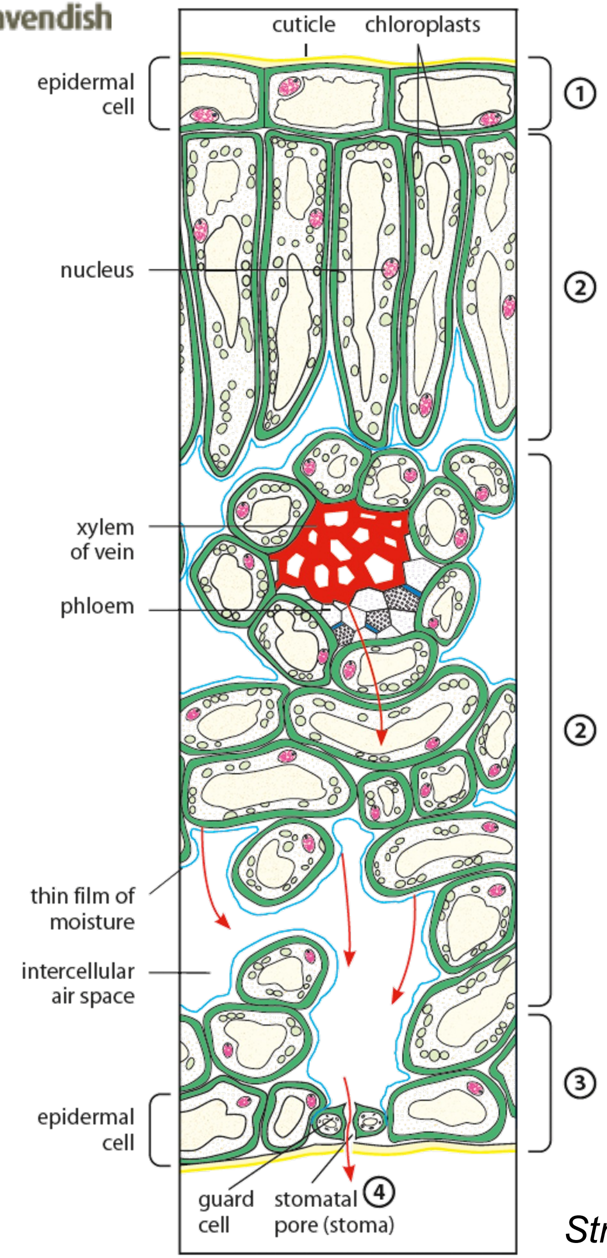
What are the external features of a leaf?



External features of a simple leaf (Figure 6.17 of Student's Book)

The leaves of grasses usually have parallel veins.

The leaves of trees and bushes usually have a network of veins.



What is the internal structure of the leaf blade?

1. The upper epidermis is made up of a single layer of closely packed cells. The cells are covered by a waxy and transparent cuticle.
2. The mesophyll is the main site of photosynthesis. It consists of palisade mesophyll and spongy mesophyll.
3. The lower epidermis consists of a single layer of closely packed cells. The cells are covered by a layer of cuticle.
4. The stomata are minute openings found in the lower epidermis.

Structure of a leaf (Figure 6.18 of Student's Book)

How is the leaf adapted for photosynthesis?

Adaptation	Function
Leaf stalk	It holds the leaf in position to absorb maximum light energy.
Thin and broad leaf blade	A thin leaf blade provides a short diffusion distance for gases and enables light to reach all mesophyll cells easily. A broad leaf blade provides a large surface area for maximum absorption of light.
Waxy cuticle on the upper and lower epidermis	It reduces water loss through evaporation from the leaf. It is transparent for light to enter the leaf.
Stomata present in the epidermal layers	Stomata open in the presence of light, allowing carbon dioxide to diffuse into and oxygen to diffuse out of the leaf.

How is the leaf adapted for photosynthesis?

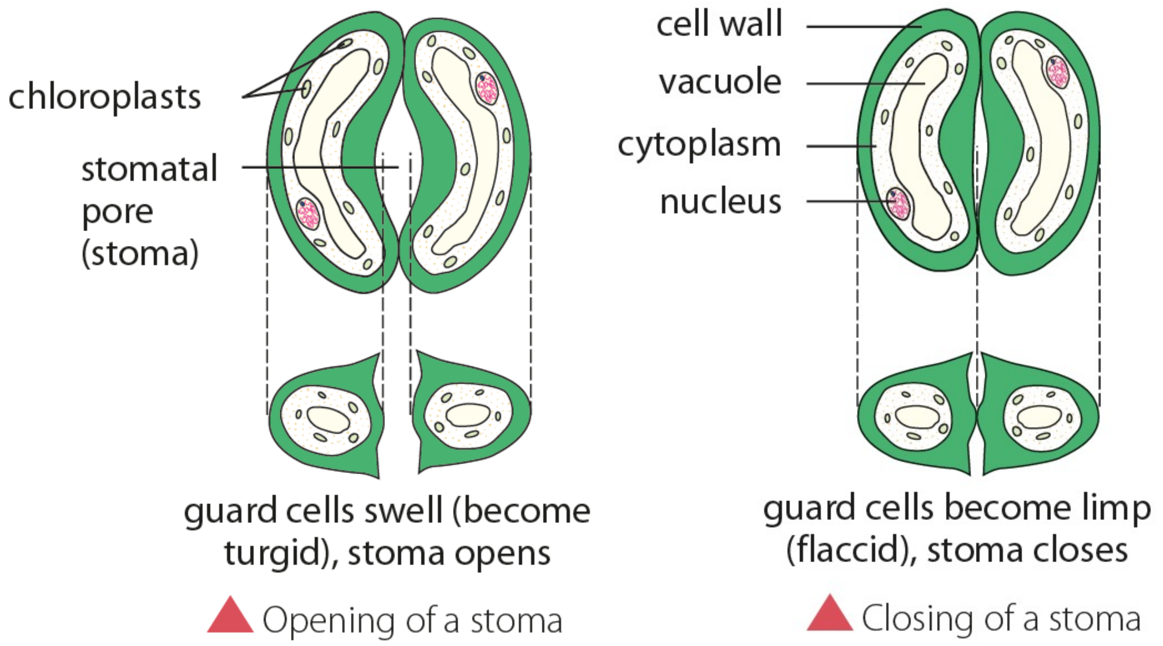
Adaptation	Function
Chloroplasts containing chlorophyll in all mesophyll cells	Chlorophyll absorbs and transforms light energy to chemical energy used in the manufacture of glucose.
More chloroplasts in upper palisade tissue	More light energy can be absorbed near the leaf surface.
Interconnecting system of air spaces in the spongy mesophyll	The air spaces allow rapid diffusion of carbon dioxide and oxygen into and out of mesophyll cells.
Veins containing xylem and phloem situated close to mesophyll cells	Xylem transports water and mineral ions to mesophyll cells. Phloem transports sugars away from the leaf.

LINK

Find out more about the xylem and phloem as transport systems in Chapter 8.

How do guard cells control the size of stomata?

- In sunlight, the guard cells manufacture glucose. They absorb water from nearby epidermal cells. They become turgid. Hence, the cells curve around the stoma and the stoma opens.

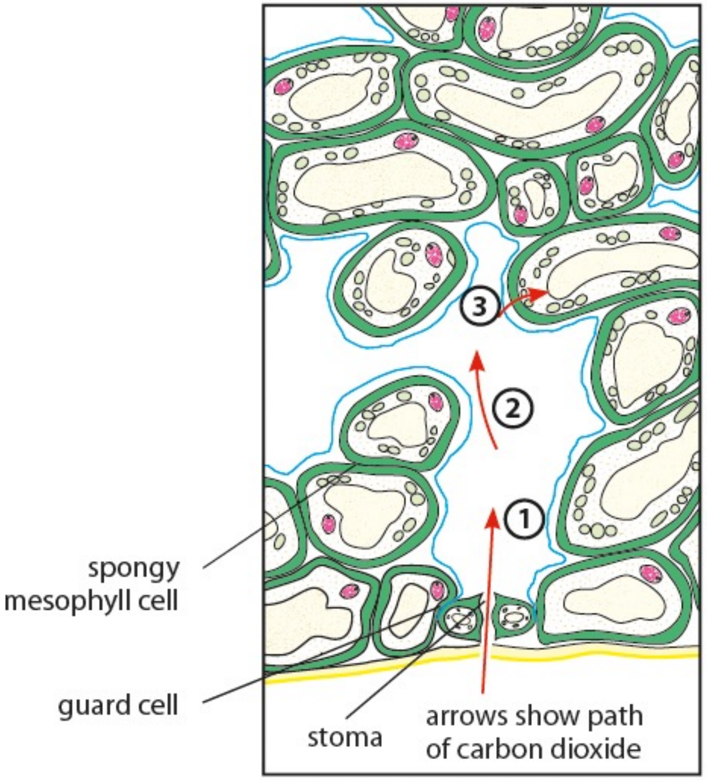


Opening and closing of a stoma (Figure 6.21 of Student's Book)

- On an extremely hot day, the excess evaporation of water causes the guard cells to become flaccid. The stoma closes. This reduces water loss by the leaf.

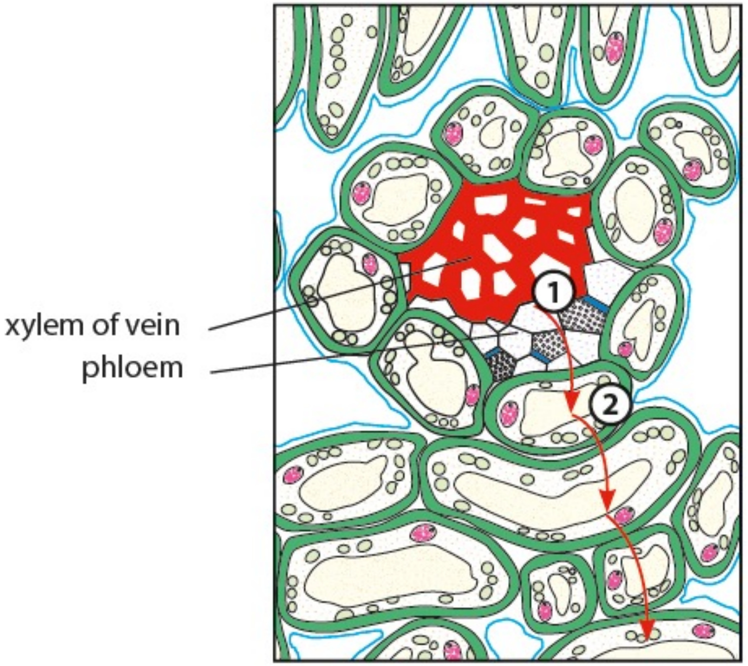
How do carbon dioxide and water enter the leaf?

- Carbon dioxide enters the leaf through the stomata.



The path of carbon dioxide into the leaf (Figure 6.23 of Student's Book)

- Xylem transports water and mineral ions to the leaf.

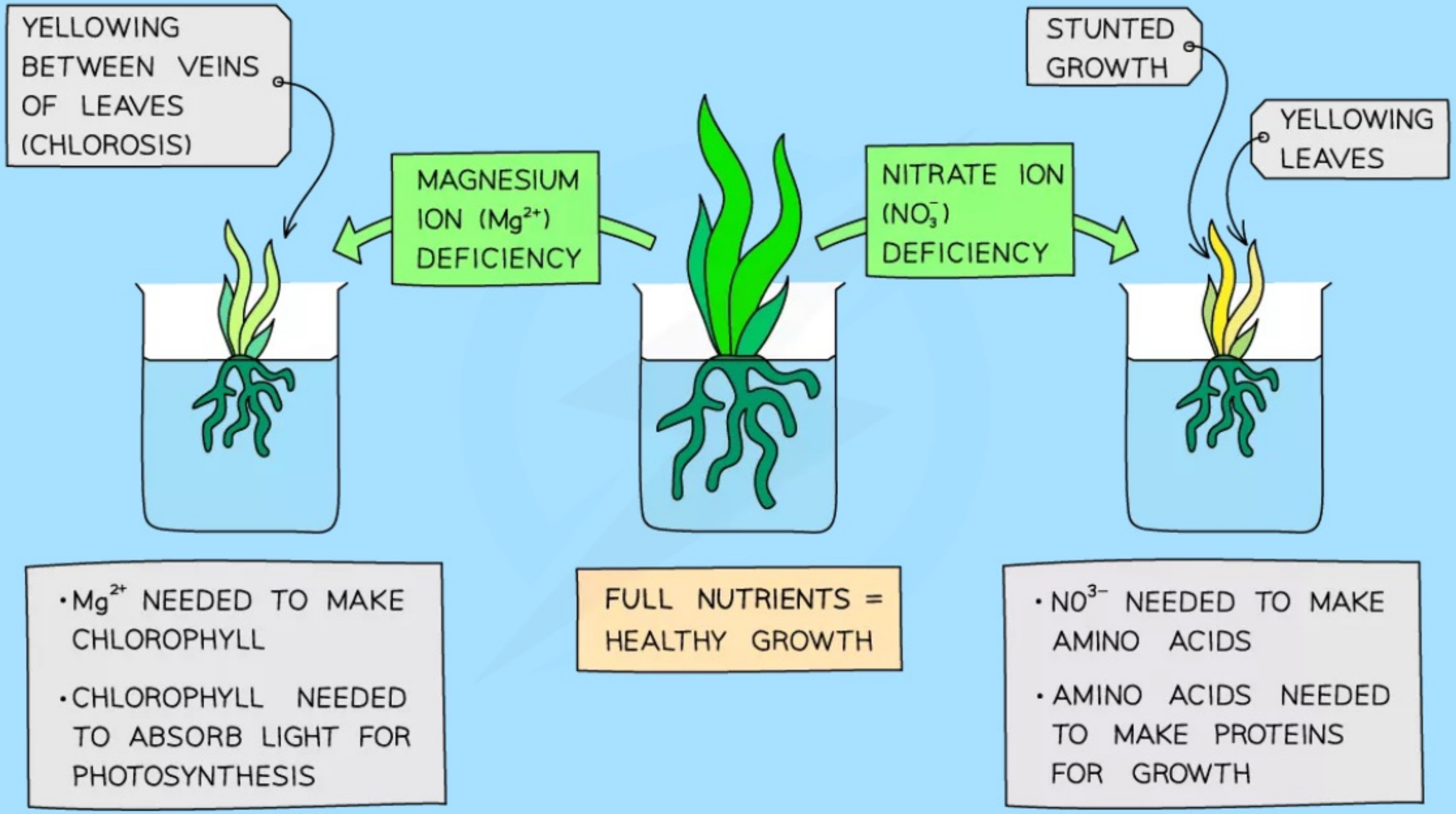


The path of water into the leaf (Figure 6.24 of Student's Book)

The Need for Mineral Ions

- Photosynthesis produces carbohydrates, but plants contain many other types of biological molecule; such as proteins, lipids and nucleic acid (DNA)
- As plants do not eat, they need to **make these substances themselves** Carbohydrates contain the elements carbon, hydrogen and oxygen but proteins, for example, contain **nitrogen** as well (and certain amino acids contain other elements too)
- Other chemicals in plants contain different elements as well, for example chlorophyll contains **magnesium** and **nitrogen**
- This means that without a source of these elements, plants cannot photosynthesise or grow properly
- Plants obtain these elements in the form of **mineral ions actively absorbed from the soil by root hair cells**
- 'Mineral' is a term used to describe any naturally occurring inorganic substance

Effects of Mineral Ion Deficiencies



MINERAL ION	FUNCTION	DEFICIENCY
MAGNESIUM	MAGNESIUM IS NEEDED TO MAKE **CHLOROPHYLL**	CAUSES YELLOWING BETWEEN THE VEINS OF LEAVES (**CHLOROSIS**)
NITRATE	NITRATES ARE A SOURCE OF NITROGEN NEEDED TO MAKE **AMINO ACIDS** (TO BUILD PROTEINS)	CAUSES **STUNTED GROWTH** AND **YELLOWING OF LEAVES**

Let's Practise 6.2

(a) Write a word equation for photosynthesis.

(b) S Write a balanced chemical equation for photosynthesis.

(c) Identify the layers labelled P, Q and R in Figure 6.26, and explain how each is adapted for photosynthesis.

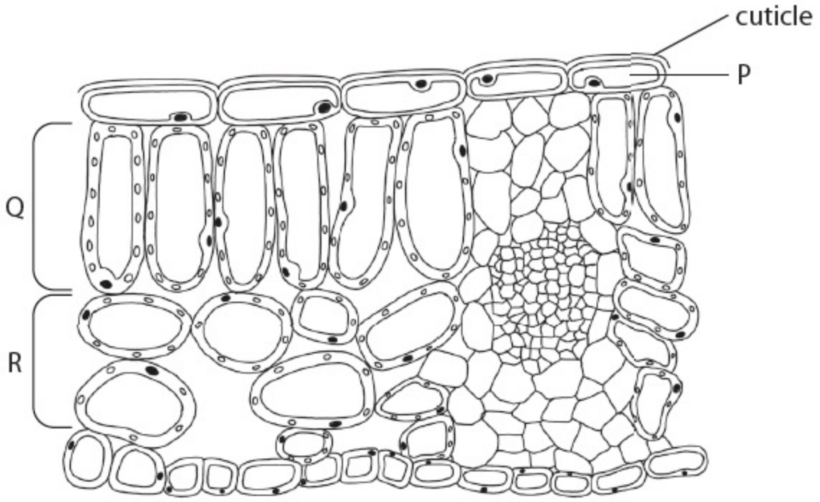
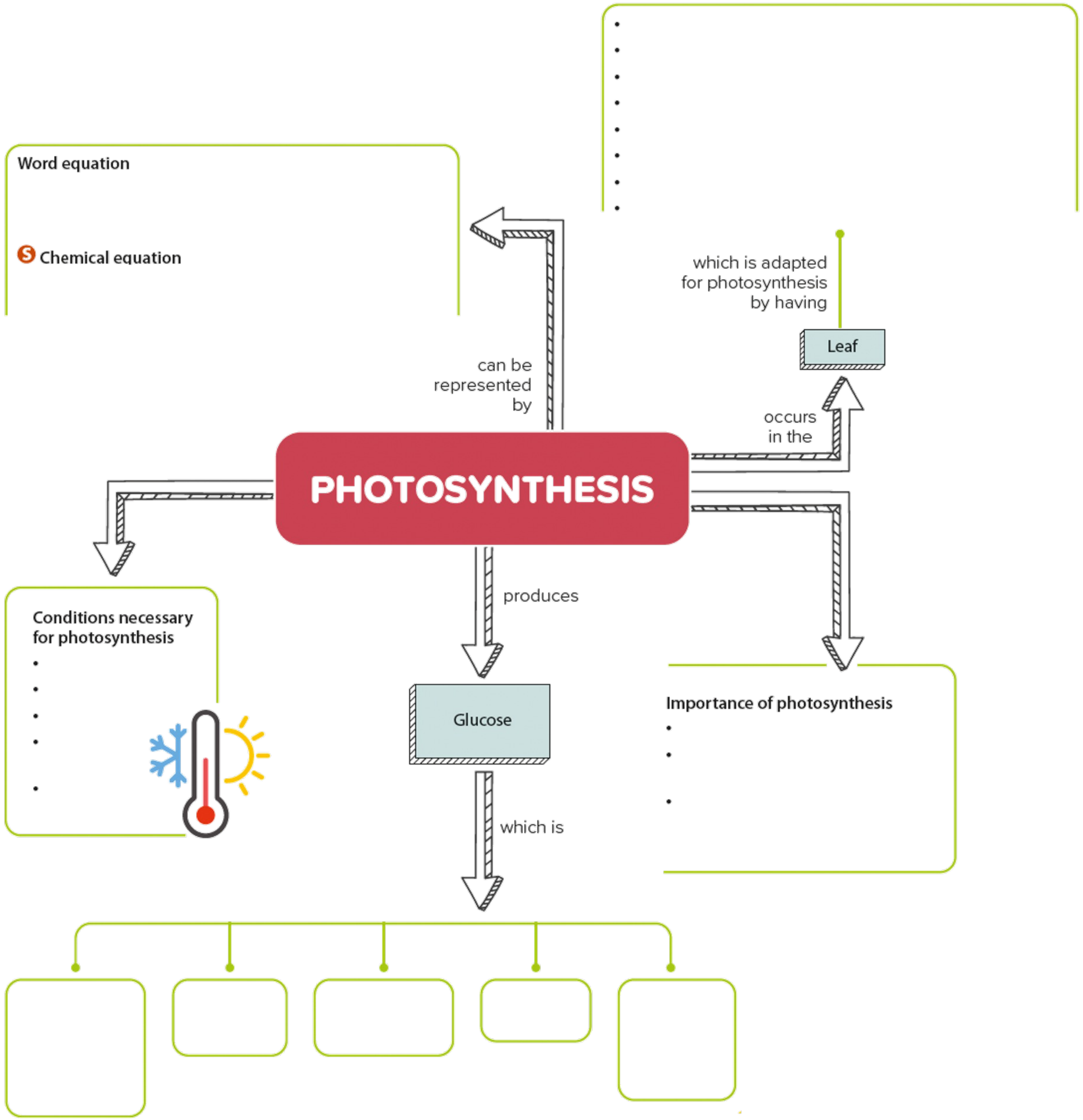


Figure 6.26 of Student's Book

What have you learnt?



What have you learnt?

Can you draw your own mind map?